

SOLAR CHALLENGER

The use of solar energy to power flight has been explored in aircraft such as the Solar Challenger. This used the Sun's rays, transformed into electricity, to boost the efforts of a pedaling pilot, giving better speed and range than purely human-powered flight. Aircraft have since been produced that are entirely solar-powered (see page 425).



improved to the point where such flights were feasible, although still extremely dangerous. The first 17 attempts to fly by balloon across the Atlantic all failed, with the loss of seven lives. Then, on August 11, 1978, Ben Abruzzo, Maxie Anderson, and Larry Newman, hailing from Albuquerque, New Mexico, set off from Presque Isle, Maine, bound for France in the helium balloon *Double Eagle II*. They named their gondola the *Spirit of Albuquerque*, in reference to Lindbergh's

Spirit of St. Louis, and took Paris' Le Bourget airport, Lindbergh's landing point, as their goal. On the evening of August 17, short of ballast and with daylight fading, they set down in a wheatfield 60 miles (96km) short of Paris. They had covered 3,100 miles (4,960km) in 137 hours 6 minutes.

Over the next two decades other records fell – the first nonstop flight across the North American continent in 1981, the first solo transatlantic balloon flight (by Joseph Kittinger) in 1984, the first solo transpacific flight (by Steve Fossett) in 1995. There remained the ultimate goal of a nonstop flight around the world.

An extremely expensive contest, in the late 1990s around-the-world ballooning turned into a battle between adventurous millionaires. American

financier Steve Fossett and British entrepreneur Richard Branson set the pace, Fossett making the first of three solo attempts in 1996. The American was one of the crew on Branson's third around-the-world attempt in 1998, which lifted off from Morocco and ended in the Pacific on Christmas Day. But these high-profile personalities were trumped by Bertrand Piccard (grandson of the inventor of the pressurized gondola) and Brian Jones, who made it around the globe in *Breitling Orbiter 3* in March 1999 (see panel, left).

Personalized flight

For many thousands of people, just flying is exciting enough, without the need for record-breaking ambitions. Gliding has developed as its own fascinating area of flight, the silent communion with the winds and the weather appealing to many aerial romantics with no taste for the roar of engines. The emergence of gliding as a skillful and satisfying sport began in German gliding clubs of the 1920s, which also provided flight training for future Luftwaffe pilots. Gliders soon proved an important testing ground for